

Judgement and Risk under Uncertainty

Behavioural Economics

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The Gambler's Fallacy

- *Gambler's fallacy*. Thinking that a departure from the average behaviour of some system will be corrected in the short term.
- Carefully note the difference between the following two questions:
 - (a) You intend to flip a fair coin eight times. What is the probability that you end up with eight heads?
 - (b) You have just flipped a fair coin seven times and ended up with seven heads. What is the probability that when you flip the coin one last time you will get another heads, meaning that you would have flipped eight heads in a row?

The Gambler's Fallacy

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- Thinking dependent outcomes are independent. A common example?

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- Stock market hedging.

Conjunction and disjunction fallacies

- “A AND B” is a conjunction; you commit the *conjunction fallacy* when you overestimate the probability of a conjunction.
- An important example: The *planning fallacy* is the mistake of making plans based on predictions that are unreasonably similar to best-case scenarios.
 - Just ask any PhD student.

Conjunction and disjunction fallacies

- There is a related fallacy called the disjunction fallacy. “A OR B” is a disjunction; you commit the *disjunction fallacy* when you underestimate the probability of a disjunction.
- The disjunction fallacy is particularly important in the context of *risk assessment*.

Are they sort of the same?

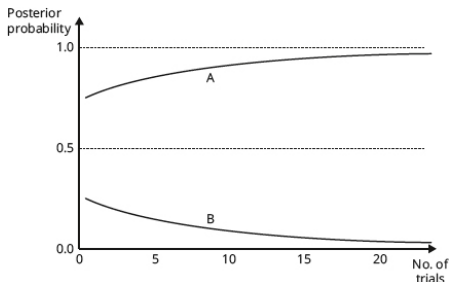
- A AND B is logically equivalent to $\neg[\neg A \vee \neg B]$.
- If you overestimate $\Pr(\neg[\neg A \vee \neg B])$, you overestimate $1 - \Pr(\neg A \vee \neg B)$.
- This is the same as underestimating $\Pr(\neg A \vee \neg B)$.
- Both the conjunction and disjunction fallacies can be explained in terms of anchoring and adjustment.

Base-rate neglect

- One source of imperfect reasoning about probabilities is confusion between conditional probabilities $\Pr(A|B)$ and $\Pr(B|A)$.
- This is important in medical tests.
- Let us look at a decision tree.

Confirmation bias

- This one is my favourite.
- People sometimes refer to this phenomenon as *washing out of the priors*.
- *Confirmation bias*. A tendency to interpret evidence as supporting prior beliefs to a greater extent than warranted.



Availability

- *Availability*. The ease with which information can be brought to mind when making a judgement.
- Different doctors react differently to similar symptoms.
- Headlines have a disproportionate effect on decision making. effect.
- Taking care of one's diet and exercise is probably more important than worrying about plane crashes and murderers.

Overconfidence

- People (including experts) express in their judgements tends to exceed the frequency with which those are correct.
- *Confidence.* statements concerning how certain you are that various things will happen.
- This leads to calibration.

Expected Utility

What is the expected value of the gamble in the St. Petersburg paradox?

$$EV(SP) = \left(\frac{1}{2}\right) 2 + \left(\frac{1}{2}\right)^2 2^2 + \left(\frac{1}{2}\right)^3 2^3 \dots = 1 + 1 + 1 \dots = \infty$$

Now consider log utility (risk averse utility function)

	H	TH	TTH	...
St Petersburg gamble	$\log(2)$	$\log(4)$	$\log(8)$	...

$$EV(SP_{\log}) = \log(2) \sum_{i=1}^{\infty} \frac{i}{2^i} = 2 \log(2) < \infty$$

*Solve using relevant formula for the AG-progression (infinite series).